Risks of Heart Disease: A MTH 345 Final Paper

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Contents

1 Abstract

In this study, I utilized a data set from Kaggle, encompassing $10,000$ observations and 21 variables. In my analysis, I will use **heart disease status** as the response variable, which is a categorical variable with two classes: *yes* and *no*. The remaining 20 variables are as follows: **Age** (discrete quantitative – years), **Gender** (categorical – male or female), **Blood pressure** (discrete quantitative – mm Hg), **Cholesterol level** (discrete quantitative – mg/dL), **Exercise habits** (categorical – high, medium, or low), **Smoking** (categorical – yes or no), **Family history of heart disease** (categorical – yes or no), **Diabetes** (categorical – yes or no), **BMI** (continuous quantitative – kg/m^2), **High blood pressure** (categorical – yes or no), **Low HDL cholesterol** (categorical – yes or no), **High LDL cholesterol** (categorical – yes or no), **Alcohol consumption** (categorical – high, medium, low, or none), **Stress level** (categorical – high, medium, or low), **Sleep hours** (continuous quantitative – hours), **Sugar consumption** (categorical – high, medium, or low), **Triglyceride level** (discrete quantitative – mg/dL), **Fasting blood sugar** (discrete quantitative – mg/dL), **CRP level** (continuous quantitative – C-reactive protein, mg/L), and **Homocysteine level** (continuous quantitative – µmol/L). I will incorporate logistic and LASSO regression, bagging, random forests, XGboost, KNN, Naive Bayes, and discriminate analysis to focus on classification and maximize model predictability on new data. For the regression component, I will address all relevant conditions for use to figure out variable significant. I checked the conditions for use for regression models, of which all were met, mitigating the need for transformations. However, I did need to do some data transformation to make the categorical variables numeric for some of the other statistical models, omitting observations with NA values, and breaking the data set into training and testing set. I then used the regression models to predict the main cause of heart disease. My analysis revealed statistically discernible evidence that BMI is predictive of heart disease status ($p-$value < 0.05). Along with the regression models, I tested accuracy, sensitivity, and specificity on all models to maximize model predictability. Quadratic Discriminate Analysis was the best model with accuracy of 79.98%, sensitivity of 0.47%, and specificity of 99.91%. This study highlights the importance of considering multiple predictors in medical diagnosis and utilizing the best statistic model in research, providing valuable insights for future research and practical applications.

2 Introduction

2.1 Background

Heart disease is the leading cause of death in the United States, claiming the lives of over 900,000 people each year—that's roughly one person every 34 seconds, or nearly 2,500 lives per day (Sager, 2025). Among the many forms of heart disease, one of the most serious and common is a myocardial infarction, more commonly known as a heart attack. A heart attack occurs when blood flow to part of the heart is severely reduced or completely blocked, usually due to a buildup of fat, cholesterol, and other substances in the coronary arteries. This blockage can damage or destroy heart muscle tissue due to a lack of oxygen-rich blood (*Heart Attack*, 2023).

There are many risk factors that increase the likelihood of having a heart attack. These include age, sex, genetic predisposition, autoimmune diseases, diabetes, high blood pressure, high cholesterol, obesity, lack of physical activity, smoking, excessive alcohol use, and chronic stress. Men are at a greater risk of heart attacks and tend to experience them earlier in life, with an average onset age of 45 compared to 55 for women(*Heart Attack*, 2023). Genetics also play a significant role—those with a family history of heart disease are more likely to suffer from heart attacks themselves.

People of color face disproportionately high rates of heart attacks, often due to higher rates of hypertension, obesity, and diabetes, as well as reduced access to quality healthcare and nutritious foods (*Understand Your Risks to Prevent a Heart Attack*, 2024). Smoking is another major contributor. Nicotine raises heart rate and blood pressure, increases the likelihood of blood clots, and promotes the buildup of plaque in arteries. Even secondhand smoke can increase the risk of heart disease in non-smokers(*Understand Your Risks to Prevent a Heart Attack*, 2024).

Other contributing factors include high blood pressure, which can damage arteries over time and make the heart work harder, leading to thickened heart muscle. Elevated levels of low-density lipoprotein (LDL) cholesterol—commonly referred to as "bad" cholesterol—can narrow arteries. High triglyceride levels, another type of blood fat, also contribute to heart attack risk(*Heart Attack*, 2023).

2.2 Data and Reason

The dataset I am using was created by Oktay Ordekci and is available on Kaggle, an online platform that hosts thousands of datasets. It includes a wide range of health indicators and risk factors associated with heart disease. Variables such as age, gender, blood pressure, cholesterol levels, smoking habits, and exercise patterns have been collected to analyze heart disease risk and support health-related research. This dataset can be utilized by healthcare professionals, researchers, and data analysts to examine trends, identify contributing factors, and perform in-depth analyses related to cardiovascular health (Ordekci, 2025).

The dataset contains 10,000 observations and 21 variables. For my research, I will use **heart disease status** as the response variable, which is a categorical variable with two classes: *yes* and *no*. The remaining 20 variables are as follows:

- **Age** (discrete quantitative – years)
- **Gender** (categorical – male or female)
- **Blood pressure** (discrete quantitative – mm Hg)
- **Cholesterol level** (discrete quantitative – mg/dL)
- **Exercise habits** (categorical – high, medium, or low)
- **Smoking** (categorical – yes or no)
- **Family history of heart disease** (categorical – yes or no)
- **Diabetes** (categorical – yes or no)
- **BMI** (continuous quantitative – kg/m^2)
- **High blood pressure** (categorical – yes or no)
- **Low HDL cholesterol** (categorical – yes or no)
- **High LDL cholesterol** (categorical – yes or no)

- **Alcohol consumption** (categorical – high, medium, low, or none)
- **Stress level** (categorical – high, medium, or low)
- **Sleep hours** (continuous quantitative – hours)
- **Sugar consumption** (categorical – high, medium, or low)
- **Triglyceride level** (discrete quantitative – mg/dL)
- **Fasting blood sugar** (discrete quantitative – mg/dL)
- **CRP level** (continuous quantitative – C-reactive protein, mg/L)
- **Homocysteine level** (continuous quantitative – µmol/L)

I am interested in this dataset because it offers a comprehensive view of the various risk factors linked to heart disease. By analyzing these variables, my goal is to identify which risk factors have the most significant impact on heart health. Understanding these relationships could help inform preventive strategies and contribute to broader health research. In my analysis, I aim to determine the most influential predictors while maximizing the accuracy of heart disease prediction.

3 EDA

3.1 Data Manipulation

Before diving into an exploration of the data, I must first prepare the data for the later tests. The original data set contains $10,000$ observations and 21 different variables. Out of the 21 variables, 12 of them are categorical variables. We are going to use Heart.Disease.Status variable as the response variable. For all the categorical variables, we will use it as a factor of the response variable, along with the rest of the variables. We will mutate them and give them a numeric value. For the gender variable, male will be 0 and female will be 1. For the categorical variables with yes and no as a response, yes is 0 and no is 1. For categorical variables with high, medium, and low as a response, high is 0, medium is 1, low is 2, and none is 3. I also removed any observations that contained missing values, leaving $9,500$ observations.

I also split this final data set into training and testing data, with 80% of the data going to training and 20% going to testing. There are $7,600$ observations in the training set and $1,900$ in the testing set. This process is crucial to creating predictive models, as it allows one to test the model's accuracy on data it hasn't seen before. Overall, I hope that this will allow me to find a model with low bias on the training data and low variability on the testing data without loss of accuracy.

3.2 Response

I will first quickly investigate the response variable: whether the date resulted in heart disease or not. As it is just a dichotomous response, with results being either "Yes" (0) to indicate heart disease or "No" (1) to indicate no heart disease, a bar graph is shown in Figure 1 of the observations. Out of the $9,500$ observations, only $1,903$ resulted in heart disease, meaning about 80% the sample resulted in not having a heart disease.

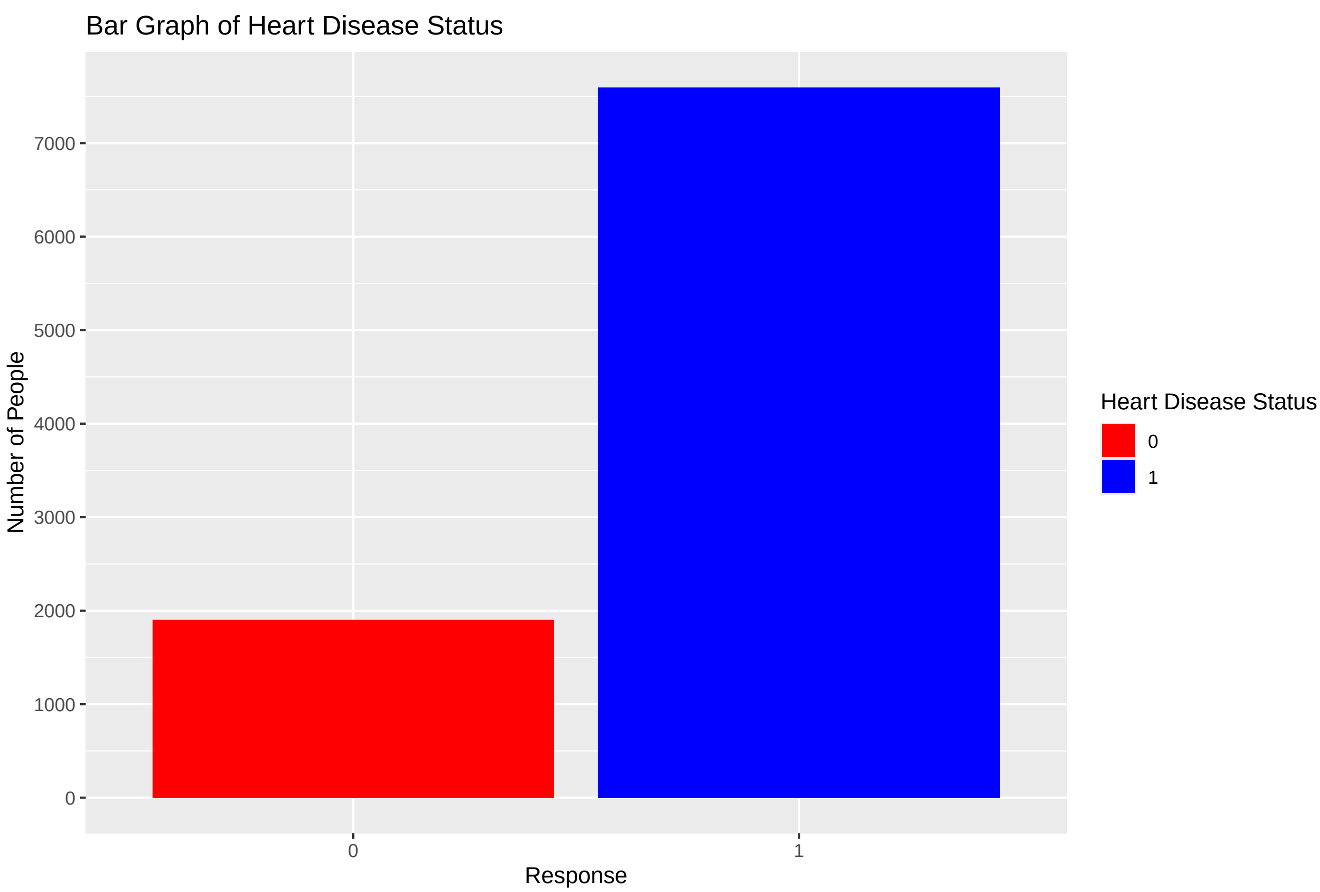

Figure 1: Bar Plot of Heart Disease Status.

3.3 Predictors

I will next explore the predictor variables and see how evenly distributed it is to see how it will affect the data.

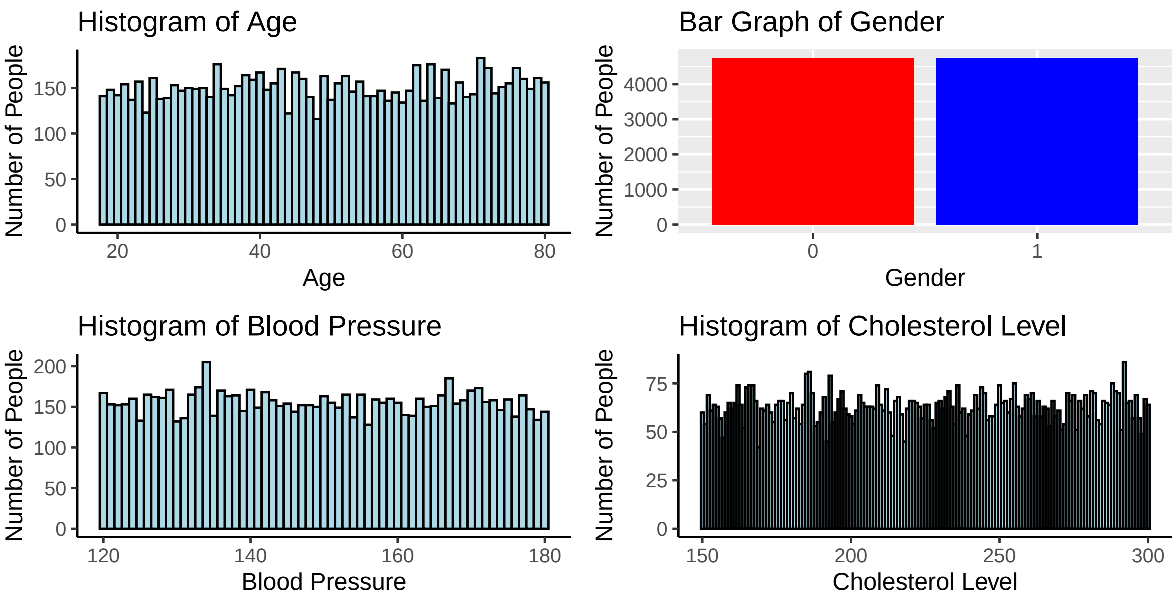

Figure 2: Graphs for Age, Gender, Blood Pressure, Cholesterol Level.

As seen in Figure 2, all the predictors are evenly distributive between Age, Gender, Blood Pressure, and Cholesterol Level. The histograms are approximately uniform and the bar graph is about even.

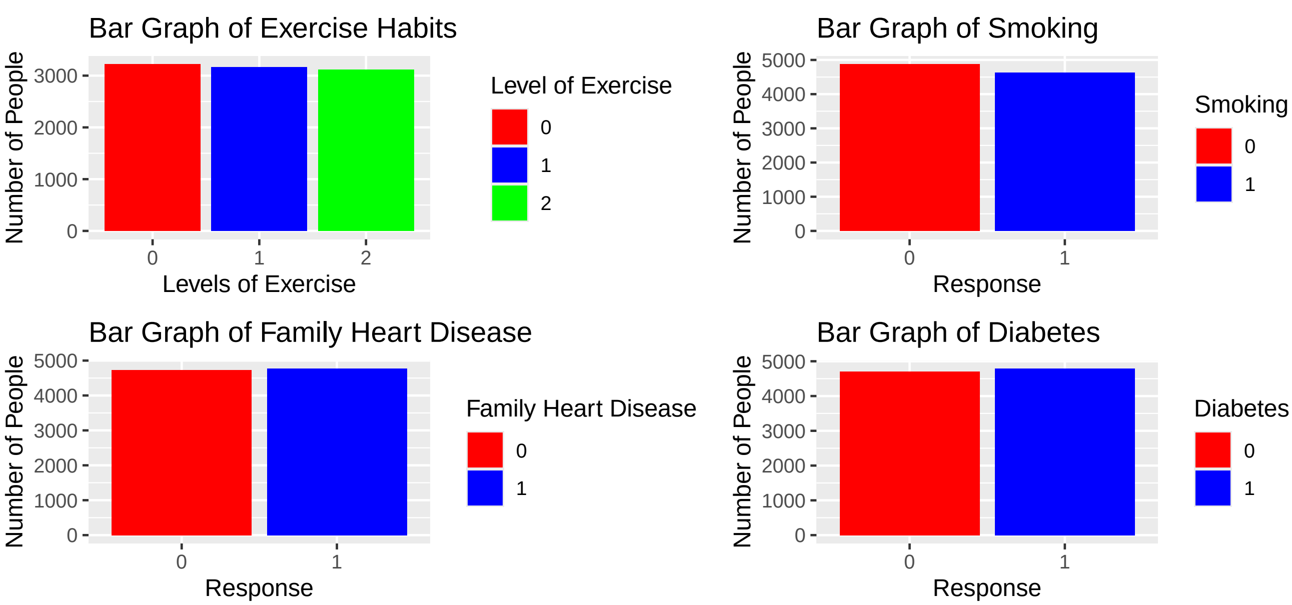

Figure 3: Graphs of Exercise Habits, Smoking, Family Heart Disease, Diabetes.

As seen in Figure 3, all the predictors are evenly distributive between Exercise Habits, Smoking, Family Heart Disease, and Diabetes. The bar graphs is about even.

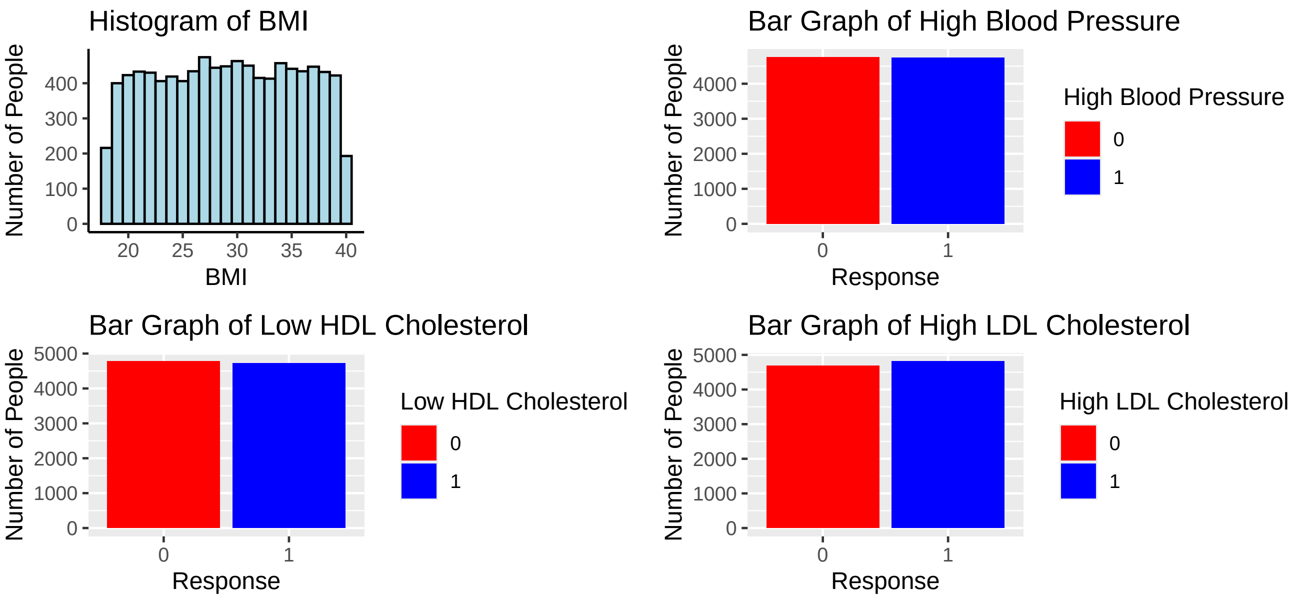

Figure 4: Graphs of BMI, High Blood Pressure, Low HDL Cholesterol, High LDL Cholesterol.

As seen in Figure 4, all the predictors are evenly distributive between BMI, High Blood Pressure, Low HDL Cholesterol, and High LDL Cholesterol. The bar graphs is about even and the histogram is around uniform.

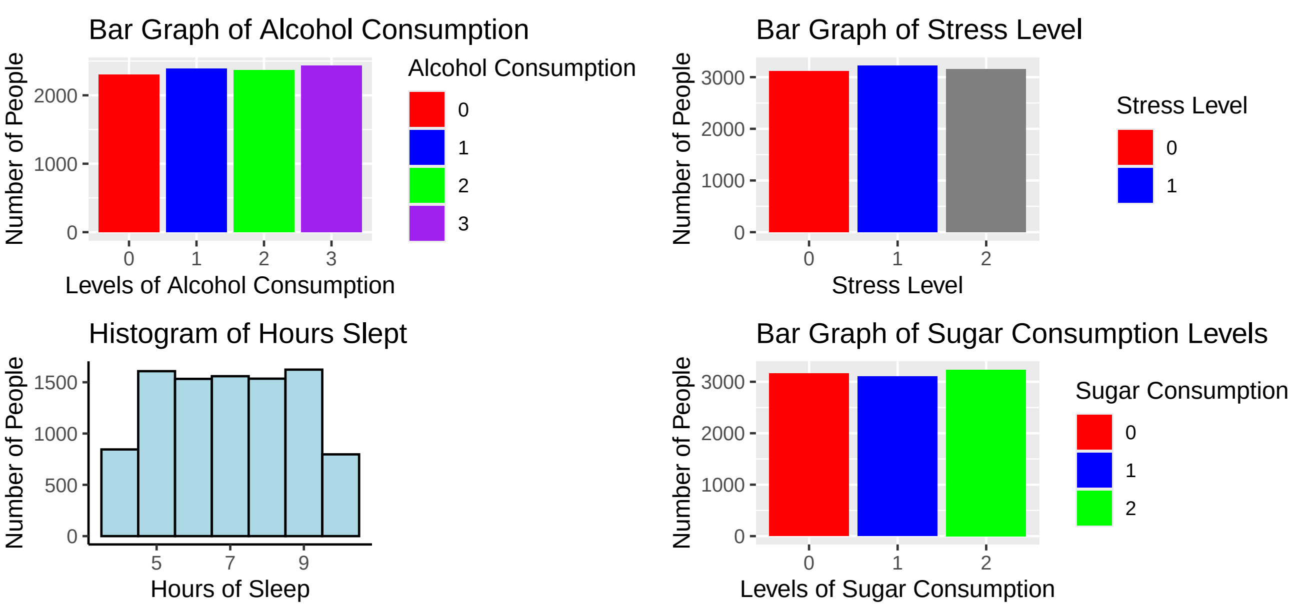

Figure 5: Graphs for Alchol Consumption, Stress Level, Hours Slept, Sugar Consumption.

As seen in Figure 5, all the predictors are evenly distributive between Alcohol Consumption, Stress Levels, Hours Slept, and Sugar Consumption. The bar graphs is about even and the histogram is around uniform.

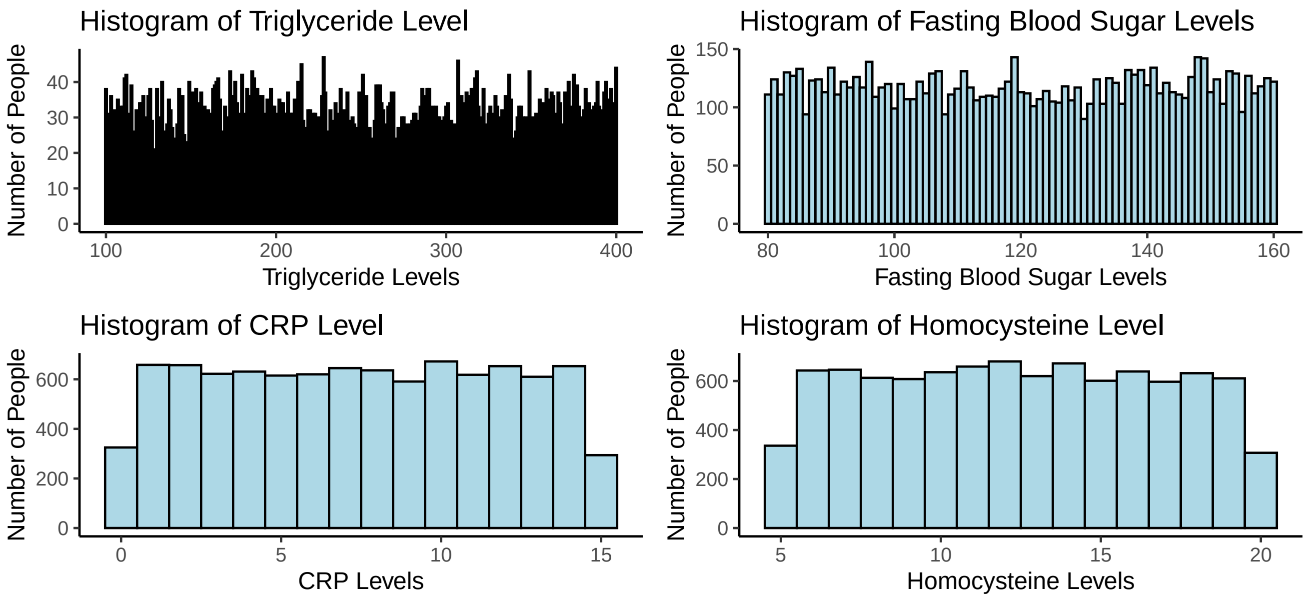

Figure 6: Graphs for Triglyceride Levels, Fasting Blood Sugar Level, CRP Levels, and Homocysteine Level.

As seen in Figure 6, all the predictors are evenly distributive between Triglyceride, Fasting Blood Sugar, CPR, and Homocysteine Levels. The histograms is around uniform.

4 Logistic Regression

4.1 Assessing Assumptions

Before I can create a logistic regression model to predict for heart disease, assumptions must first be assessed. The response variable is dichotomous, with an outcome of either "Yes", indicating heart disease, or "No", indicating no heart disease. The data did not specify where the responses came from. Therefore, it is difficult to tell what population the sample could be representative of. Additionally, there is no indication of how the samples were collected, meaning I cannot be sure that these are all independent observations. If the results were taken randomly from some individuals that participated, then the observations would be. However, I will assume that the observations are independent as heart disease is not contagious, so one person having it doesn't affect the sample by spreading it to others. Also the risk factors are based on genetics and the person lifestyle.

4.2 LASSO

Lasso Regression is a valuable tool for handling high-dimensional data and performing automatic feature selection. By introducing a penalty term, it enhances model simplicity and interpretability while preventing overfitting. It selects important features by shrinking less significant ones to zero.

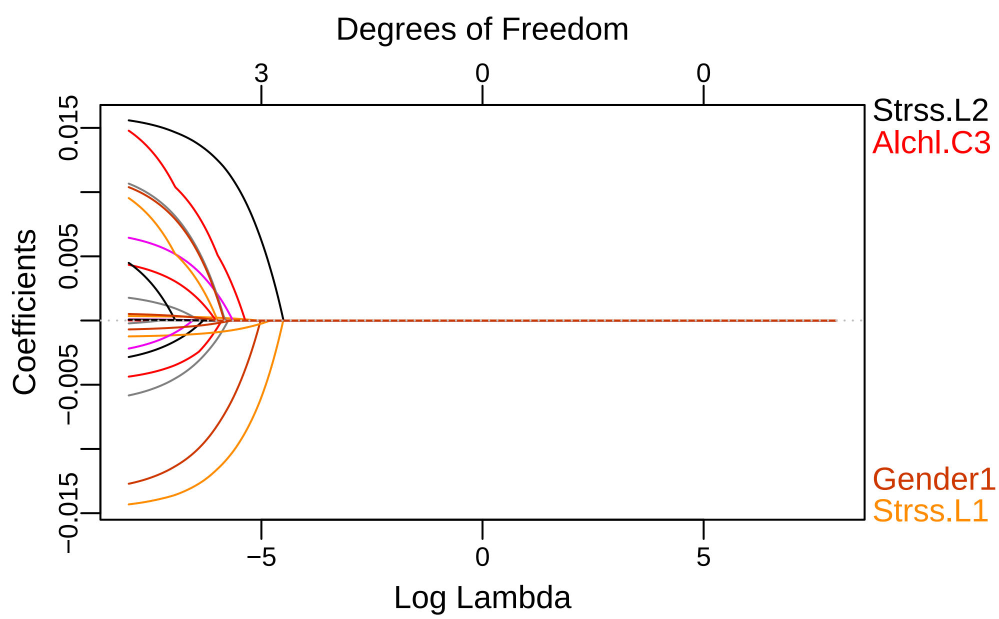

Figure 7: Comparing log lambda to the LASSO regression coefficients.

According to Figure 7, the biggest factors are alcohol consumption, gender, and stress levels.

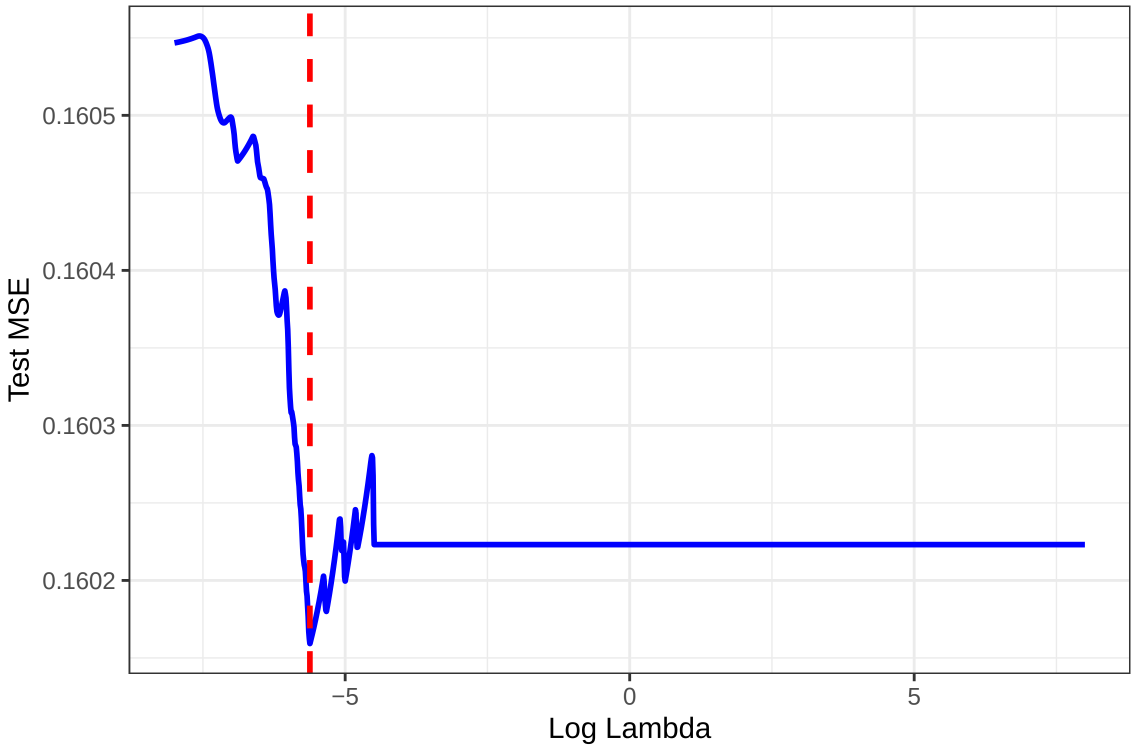

Figure 8: Cross-validation curve for LASSO regression; log lambda versus test MSE using LOOCV. The red dashed line represents the optimal log lambda value.

According to Figure 8, the data is center around log lambda of -5.62 with the smallest mse of 0.1601593.

Table 1: Lasso Regression Coefficients

Variable	**Coefficient**
(Intercept)	1.8002688681
Age	—
Gender1	-0.0058844390
Blood Pressure	0.0001520150
Cholesterol Level	—
Exercise Habits1	—
Exercise Habits2	—
Smoking1	—
Family Heart Disease1	—
Diabetes1	—
BMI	-0.0007305023
High Blood Pressure1	—
Low HDL Cholesterol1	—
High LDL Cholesterol1	—
Alcohol Consumption1	—
Alcohol Consumption2	—
Alcohol Consumption3	0.0024509254
Stress Level1	-0.0101011258
Stress Level2	0.0108574110
Sleep Hours	—
Sugar Consumption1	—
Sugar Consumption2	—
Triglyceride Level	—
Fasting Blood Sugar	—
CRP Level	—
Homocysteine Level	—

As shown in Table 1, most coefficients are zero. In the case of LASSO Regression, the variables that go to zero is the least significant features in predicting heart disease. Besides the intercept (indicating heart disease), there are 6 significant predictors: gender 1, blood pressure, BMI, Alcohol Consumption 3, and Stress Level 1 and 2.

Prediction	0	1
0	0	0
1	1903	7597

Table 2: Confusion Matrix For LASSO Regression

Table 2 is a confusion matrix, to predict accuracy of LASSO Regression. The accuracy is 79.9684 %, with sensitivity of 0 and specificity of 1.

4.3 Full Predictor Models

The data seem to not be correlated, but I will further check this later by looking at the variance inflation factors of the logistic regression models. Similarly, linearity will be assessed later using the Box-Tidwell test. The first model I will fit in order to predict for matches is a logistic regression model using all the predictors. I will first look at multicollinearity through the VIF values.

Table 3: Variance Inflation Factors of the predictors in the full predictor logistic regression model.

Variable	**VIF**
Age	1.0026
Gender	1.0025
Blood Pressure	1.0019
Cholesterol Level	1.0026
Exercise Habits	1.0032
Smoking	1.0021
Family Heart Disease	1.0016
Diabetes	1.0023
BMI	1.0028
High Blood Pressure	1.0023
Low HDL Cholesterol	1.0021
High LDL Cholesterol	1.0028
Alcohol Consumption	1.0081
Stress Level	1.0049
Sleep Hours	1.0022
Sugar Consumption	1.0055
Triglyceride Level	1.0017
Fasting Blood Sugar	1.0015
CRP Level	1.0028
Homocysteine Level	1.0023

As seen in Table 3, there is no concern of multicollinearity as all of the features have a VIF around 1. Finally, I will perform the Box-Tidwell test to assess linearity through logistic regression. As you can see with the RQR plot below, there is no need for transformations of generalized linear model as there is no pattern. The plot is just a blob.

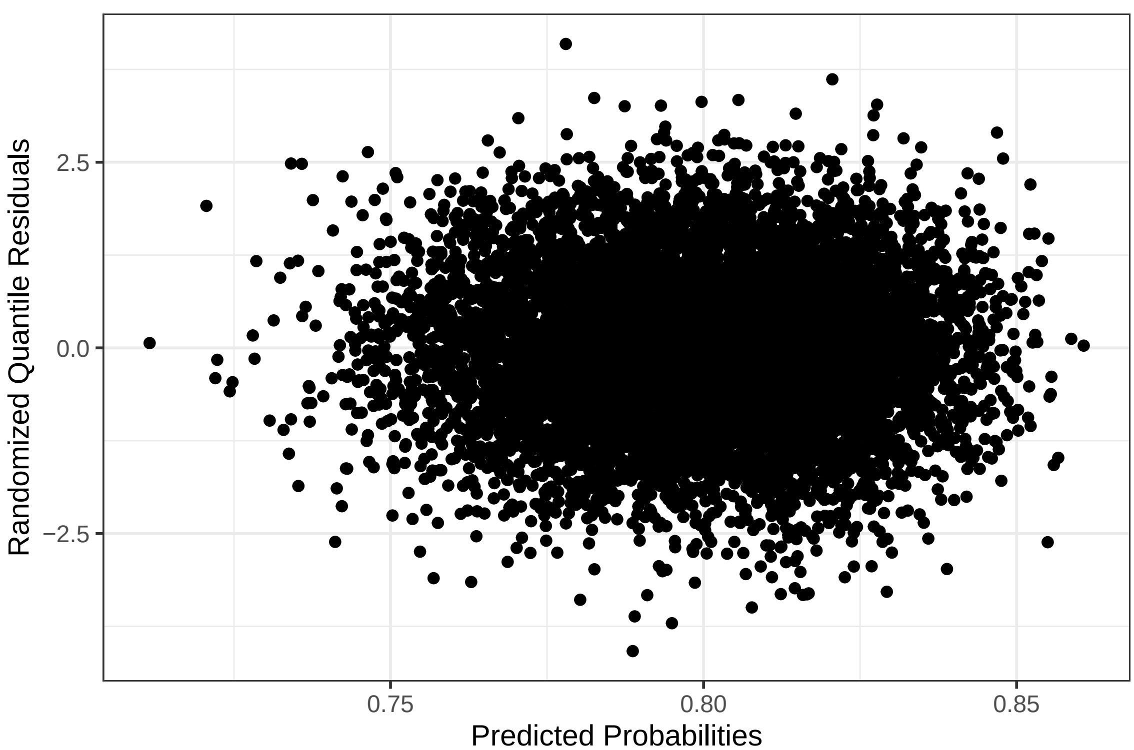

Figure 9: A plot of the predicted probabilites versus the randomized quantile residuals for Logistic Regression.

Now, that we have test all the assumptions for logistic regression and narrow down the number of significant predictors with the LASSO Regression, now it is time to use logistic regression to find the number one predictor of heart disease. The table below will show that.

Table 4: Coefficients of logistic regression model.

Variable	**P-value**
(Intercept)	0.0018
Age	0.6426
Gender1	0.1020
Blood Pressure	0.1161
Cholesterol Level	0.8716
Exercise Habits1	0.6283
Exercise Habits2	0.8237
Smoking1	0.5419
Family Heart Disease1	0.4269
Diabetes1	0.6685
BMI	0.0489
High Blood Pressure1	0.9913
Low HDL Cholesterol1	0.7327
High LDL Cholesterol1	0.3875
Alcohol Consumption1	0.3115
Alcohol Consumption2	0.5607
Alcohol Consumption3	0.1407
Stress Level1	0.1496
Stress Level2	0.1062
Sleep Hours	0.8567
Sugar Consumption1	0.2376
Sugar Consumption2	0.2439
Triglyceride Level	0.9948
Fasting Blood Sugar	0.6357
CRP Level	0.5304
Homocysteine Level	0.4205

As seen in Table 4, the one significant predictor of heart disease is BMI. That is because it is the only feature with a p-value less than 0.05, besides the intercept. Next, I will make a confusion matrix to calculate accuracy, specificity, and sensitivity.

Prediction	0	1
0	0	0
1	1903	7597

Table 5: Confusion Matrix For Logistic Regression

Table 5 is a confusion matrix to predict accuracy of Logistic Regression. The accuracy is 79.9684 %, with sensitivity of 0 and specificity of 1. With specificity and sensitivity not being so good, we get the figure below.

5 Classification Trees

5.1 Base Tree

The next method I will explore is the dichotomous classification tree. I decided to first create a base tree just using all predictors, but fitting it on the training set to make predictions on the testing set. However, the base tree ended up being just the root node. In other words, the base classification tree always predicted for an outcome of no heart disease, similar to the logistic regression model and the intercept model after performing LASSO Regression. Therefore, confusion matrices would again be similar to that in Logistic Regression, with the same specificity of 1 and sensitivity of 0, but slightly lower accuracy of 79.95 , as seen below in Table 6. As this method does not perform any better than previous ones, I will move on to other tree-based algorithms that could help improve the predictability of the data.

Prediction	0	1
0	0	0
1	381	1519

Table 6: Confusion Matrix For Decision Tree

5.2 Bagging

I next decided to perform bagging on the classification tree. While this algorithm greatly decreases interpretability by combining a large number of trees together and taking the outcome with the majority of votes, it usually reduces the variabilityof individual models, making the overall model more robust, improving reliability. It also avoids overfitting as the process of creating diverse subsets of the data ensures that each model is trained on different parts of the dataset. The final predictions are aggregated through methods like majority voting for classification or averaging for regression, which enhances overall performance.

Prediction	0	1
0	5	23
1	376	1496

Table 7: Confusion Matrix For Bagging

As seen in Table 7, the confusion matrix has an accuracy of 79.00%, specificity of 98.49%, and sensitvitiy of 1.31 %.

5.3 Random Forest

Random Forest creates an ensemble of multiple decision trees, where each tree is trained on a random subset of the data and a random subset of features. This diversity among the trees helps to reduce overfitting and improve the overall performance of the model. Random Forests are versatile and can be used for both classification and regression tasks.

Prediction	0	1
0	0	0
1	381	1519

Table 8: Confusion Matrix For Random Forest

Based on Table 8, the accuracy of Random Forest is 97.95 % with specificity of 1 and sensitivity of 0.

5.4 Boosting

XGBoost is an extreme gradient boosting algorithm, incorporating regularization techniques to prevent overfitting and improve performance. It is known for its speed and efficiency, handling both numerical and categorical features effectively. XGBoost uses a regularized objective function, which includes both the loss function and a regularization term. I used it instead of AdaBoost because it is faster and more convenient.

Prediction	0	1
0	0	0
1	381	1519

Table 9: Confusion Matrix For Boosting

Based on Table 9, the accuracy of XGBoosting is 97.95 % with specificity of 1 and sensitivity of 0.

6 Other Prediction Techniques

6.1 K-Nearest Neighbors

KNN identifies the K-nearest neighbors to a given data point using a distance metric, making predictions based on similar data points in the dataset. Euclidean distance is the shortest distance between two points in a metric space. It is calculated as the square root of the sum of the squared differences between the corresponding coordinates of the two points. Manhattan distance measures the distance between two points by summing the absolute differences of their coordinates. Even tough, I did the 2 different KNN tests, I got the same results as seen below in Table 10 . The accuracy of KNN Manhattan and Euclidean is 97.95 % with specificity of 1 and sensitivity of 0.

Prediction	0	1
0	0	0
1	381	1519

Table 10: Confusion Matrix For KNN Manhattan and Euclidean

6.2 Naive Bayes

Naive Bayes is used for cassification tasks, based on Bayes' Theorem to find probabilities of different classes given the features of the data.

Prediction	0	1
0	0	0
1	381	1519

Table 11: Confusion Matrix For Naive Bayes

Based on Table 11, the accuracy of Naive Bayes is 97.95 % with specificity of 1 and sensitivity of 0.

6.3 Discriminant Analysis

Discriminant analysis is a statistical technique used to classify observations into predefined classes. By analyzing variances and covariances, discriminant analysis finds linear combinations of features that best separate the groups, maximizing differences between groups while minimizing differences within groups. The 2 main types of discriminant analysis is Linear and Quatractic. LDA is the most commonly used form of discriminant analysis, suitable when the predictor variables are continuous and follow a normal distribution. It assumes that each group has identical variance-covariance matrices. It maximizes the separation between two or more groups by finding linear combinations of predictor variables that best distinguish them. QDA is similar to LDA but allows for distinct variance-covariance matrices for each group, making it more flexible in cases where groups do not have identical covariances. QDA provides better classification accuracy when groups differ in variance-covariance structures. QDA is suitable for complex classification tasks where groups have distinct distributions.

Prediction	0	1
0	0	0
1	1903	7597

Table 12: Confusion Matrix For LDA

Table 12 is a confusion matrix to predict accuracy of LDA. The accuracy is 79.9684 %, with sensitivity of 0 and specificity of 1.

Prediction	0	1
0	9	7
1	1894	7590

Table 13: Confusion Matrix For QDA

Table 13 is a confusion matrix to predict accuracy of QDA. The accuracy is 79.9989 %, with sensitivity of 0.0047 and specificity of 0.9991.

7 Conclusion

This data analysis aimed to find a model that was highly predictive of whether a person would have heart disease. The 20 top risk factors are age, gender, blood pressure, cholesterol level, exercise habits, smoking, family history of heart disease, diabetes, BMI, high blood pressure, low HDL cholesterol, high LDL cholesterol, alcohol consumption, stress level, sleep hours, sugar consumption, triglyceride level, fasting blood sugar, CRP level, and homocysteine level. My goal was to focus on classification and maximize model predictability on new data, while testing for most significant predictors.

My exploratory data analysis revealed that all variables' responses are equally distributed. The regression model met all necessary conditions for use, and further transformations were deemed unnecessary. However, I did need to do some data transformation to make the categorical variables numeric for some of the other statistical models, omitting observations with NA values, and breaking the data set into training and testing set. I then used regression models to predict the main cause of heart disease. My analysis revealed statistically discernible evidence that BMI is predictive of heart disease status ($p-\text{value} < 0.05$).

Along with the regression models, I tested accuracy, sensitivity, and specificity on all classification and discriminatory models to maximize model predictability. Quadratic Discriminate Analysis was the best model with accuracy of 79.98%, sensitivity of 0.47 %, and specificity of 99.91%. Then, the logistic models with accuracy of 79.97%, sensitivity of 0%, and specificity of 100%. Then, the decisions trees with accuracy of 79.95%, sensitivity of 0%, and specificity of 100%. The worst model was bagging with accuracy of 79%, sensitivity of 1.31%, and specificity of 98.49%.

In conclusion, I found the regression algorithms to perform best at predicting for heart disease, with BMI as the most significant predictor. Our study highlights the importance of considering multiple predictors when modeling risk factors of heart disease. The findings provide valuable insights into the health risk factors influencing heart disease and can inform future research and practical applications medical health science.

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